

Neural Inference as the Operating System on Minimal RISC-V

Advanced Systems Engineering (ASE2026)

Paris Lodron University of Salzburg

Author: Bernhard Guillon

Date: 2026-06-24

Problem definition

- ▶ Can a minimal system use neural inference as its primary computation model, replacing traditional OS and application code?

What an operating system does

- ▶ Abstracts the hardware
- ▶ Memory management
- ▶ I/O handling
- ▶ Allows multiple processes
- ▶ Multiple users
- ▶ Tries hard to separate processes and users from each other

Approach

- ▶ RISC-V + Tensor OPs
- ▶ Emulator + RTL (system verilog)
- ▶ Assembler for the extensions
- ▶ Minimalistic bootloader/runner
- ▶ Pytorch for training
- ▶ A toolchain to create runnable networks

Divide and conquer

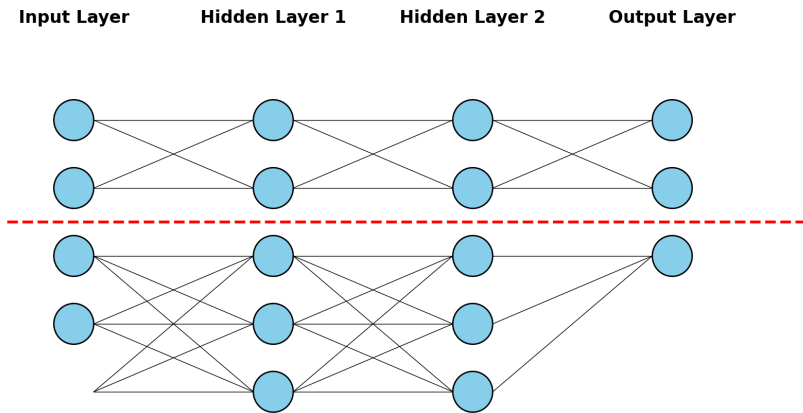
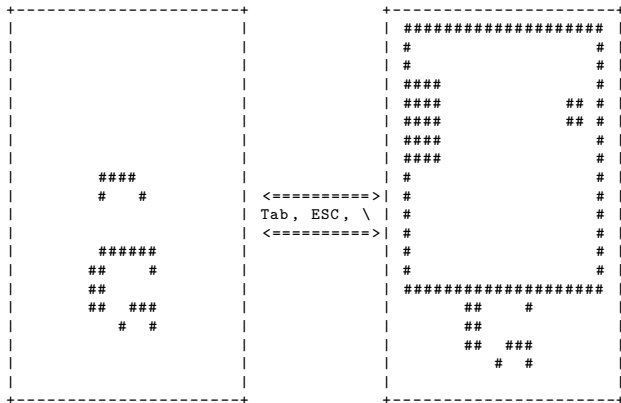


Figure 1: mlp network

Toolchain

- ▶ describe the problem
- ▶ subdivide the networks
- ▶ train the networks
- ▶ create combined model

Parallel vs time sliced execution



Findings

- ▶ Combination of MLPs
- ▶ I/O handling
- ▶ Parallel vs time sliced execution
- ▶ Virtual Hardware
- ▶ limited process and memory isolation

So long, and thanks for all the fish

- ▶ Thank you for your attention
- ▶ Feel free to ask me anything